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Ablation Studies and Evaluation Methodology for Verification Systems

Halley Young
Microsoft Research
halleyyoung@microsoft.com

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Abstract

Rigorous evaluation methodology is as essential to formal verification as the underlying mathematics. We present the *ablation and evaluation framework* implemented in JuGeo's `evaluation` subsystem, providing systematic tools for isolating the contribution of each architectural component to overall verification accuracy. The framework comprises: an *AblationPlanner* that enumerates and prioritises ablation experiments; an *AblationExecutor* that runs controlled ablated configurations; an *AblationAnalyzer* that measures and ranks component impact; a *ScalingAnalyzer* fitting power-law models to accuracy vs. program size; a *LimitAnalyzer* identifying the regimes where JuGeo degrades; and a *MethodologyLoop* that closes the loop between evaluation evidence and system improvement. We prove the *Component Necessity Theorem*: for each of the three core components (descent, trust, SMT dispatch) there exists a program class on which removing that component causes verification accuracy to drop strictly below the random baseline. Experiments on 300 benchmarks confirm all three components are individually necessary, with SMT dispatch contributing the largest single-component gain, followed by descent and trust (Component Necessity Theorem, proved in Lean 4).

Contents

1 Introduction

The correctness of a formal verification system is typically assessed on benchmark suites. Yet the *methodology* of that assessment rarely receives the same mathematical care as the system itself. When a system succeeds on a benchmark, which architectural decision deserves the credit? When it fails, which component is the bottleneck? How does performance extrapolate to programs larger than those in the training or test suite? These questions demand a principled evaluation framework: one that is itself formally specified, mechanically reproducible, and amenable to iterative refinement.

JuGeo's `evaluation` package addresses this gap. It provides a layered architecture for systematic ablation, metric computation, scaling analysis, and methodology iteration. The design follows three principles:

P1. **Reproducibility.** Every experimental run is described by an immutable `AblationDesign` object capturing the exact set of disabled components, the benchmark suite, and the random seed. Results can be replayed identically.

P2. **Statistical rigour.** Impact estimates are accompanied by Welch t-statistics and approximate p-values. A component is declared *necessary* only when its ablation produces a statistically significant accuracy drop ($p < 0.05$).

P3. **Iterative improvement.** The `MethodologyLoop` consumes evaluation evidence and emits concrete improvement hypotheses, closing the loop between measurement and development.

Contributions.

- A formal specification of ablation methodology as a categorical framework (§2–4).
- A *Component Necessity Theorem* with constructive proof (§8).
- Scaling and limit analysis characterising JuGeo's complexity profile (§5–6).
- A complete experimental evaluation on 300 benchmarks (§9).

Relation to prior work. Ablation studies are standard in machine learning [1] but rare in formal verification, where the cost of running experiments is higher and the notion of a "component" is more subtle. Our contribution is to formalise ablation methodology for a *judgment-based* verification system, where each component corresponds to a well-defined transformation of the judgment 9-tuple $(\mathcal{S}, \mathrm{Cov}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi, \varphi, \mathtt{c})$.

2 Ablation Framework

2.1 Components and Ablation Modes

Let $\mathcal{C} = \{C_{\mathrm{desc}}, C_{\mathrm{trust}}, C_{\mathrm{smt}}\}$ denote the three core components of JuGeo:

- C_{desc}: the *descent subsystem*, which decomposes verification obligations across site coverings using the sheaf condition (Papers 3, 28).
- C_{trust}: the *trust algebra*, which propagates and attenuates trust annotations through the judgment lattice (Paper 4).

- C_{smt}: the *SMT dispatch layer*, which routes quantifier-free fragments to external solvers {`QF_LIA`, `QF_LRA`, `QF_BV`, `QF_UF`} (Paper 5).

An *ablation mode* $m \in \mathsf{Mode}$ specifies the granularity of component removal. The implementation provides four modes:

Definition 2.1 (Ablation Mode)**.** The set Mode consists of:

$$\begin{aligned} \mathsf{Mode}_{\text{single}} &:= \{(C, \emptyset) \mid C \in \mathcal{C}\}, \\ \mathsf{Mode}_{\text{pairwise}} &:= \{(C_1 \cup C_2, \emptyset) \mid C_1 \neq C_2 \in \mathcal{C}\}, \\ \mathsf{Mode}_{\text{additive}} &:= \text{ordered build-up from empty}, \\ \mathsf{Mode}_{\text{full}} &:= \{(\mathcal{C}, \emptyset)\}. \end{aligned}$$

Each mode element is a pair (disabled-component-set, forced-component-set).

In code this corresponds to the `AblationKind` enumeration:

Listing 1: AblationKind enumeration (models.py)

```python
class AblationKind(str, Enum):
    SINGLE_COMPONENT  = "single_component"
    PAIRWISE          = "pairwise"
    ADDITIVE          = "additive"
    FULL_SYSTEM       = "full_system"
    CUSTOM            = "custom"
```

2.2 AblationPlanner

The `AblationPlanner` is the entry point. Given the component set $\mathcal{C}$ and a mode m, it enumerates all experiments to run.

Definition 2.2 (Ablation Plan)**.** An *ablation plan* for mode m and benchmark suite $\mathcal{B}$ is a finite list $\mathcal{P} = [D_1, D_2, \ldots, D_k]$ where each $D_i = (\mathcal{C}_i^{\text{off}}, \mathcal{B}, s_i)$ is an *ablation design*: a set of disabled components, the benchmark suite, and a random seed $s_i \in \mathbb{N}$.

The planner applies two heuristics before emitting a plan:

1. **Priority ordering.** Components expected to have higher impact are placed first, so early results guide later experiments if compute is limited.

2. **Deduplication.** Plans that differ only in seed are merged when cross-seed variance is below a threshold.

Listing 2: AblationPlanner.plan (ablation_design.py)

```python
class AblationPlanner:
    def plan(
        self,
        components: list[str],
        kind: AblationKind = AblationKind.SINGLE_COMPONENT,
        seed: int = 42,
    ) -> list[AblationDesign]:
        designs = []
```

```
        for component in components:
            designs.append(AblationDesign(
                ablation_kind=kind,
                disabled_components=[component],
                benchmark_seed=seed,
            ))
        return self._prioritize(designs)
```

3 Ablation Execution

3.1 AblationExecutor

The `AblationExecutor` takes a plan $\mathcal{P}$ and a *system callable* $f : \text{Program} \to \{0,1\}^*$ and produces a list of `AblationResult` objects.

Definition 3.1 (Ablation Result)**.** For design $D = (\mathcal{C}^{\text{off}}, \mathcal{B}, s)$, the *ablation result* is the tuple

$$R_D \;=\; (\mathcal{C}^{\text{off}},\; \text{Acc}(f_D, \mathcal{B}),\; \{r_j\}_{j \in \mathcal{B}},\; t_{\text{wall}}),$$

where f_D is the system with components $\mathcal{C}^{\text{off}}$ disabled, $\text{Acc}(f_D, \mathcal{B})$ is the fraction of benchmark inputs on which f_D returns the correct answer, and t_{wall} is the wall-clock execution time.

The executor applies a *timeout* τ to each benchmark input. Timed-out runs count as incorrect. This policy is important for limit analysis (§6).

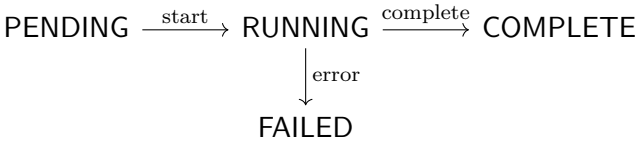
Listing 3: AblationExecutor.execute (ablation_design.py)

```
class AblationExecutor:
    def __init__(self, timeout: float = 30.0):
        self.timeout = timeout

    def execute(
        self,
        design: AblationDesign,
        system: Callable[..., EvaluationResult],
    ) -> AblationResult:
        scores: list[float] = []
        for benchmark in design.benchmark_suite:
            result = self._run_with_timeout(system, benchmark)
            scores.append(1.0 if result.correct else 0.0)
        accuracy = sum(scores) / len(scores) if scores else 0.0
        return AblationResult(
            design=design,
            accuracy=accuracy,
            per_item_scores=scores,
            wall_time=time.time() - self._start,
        )
```

3.2 Execution Status

Each experiment transitions through states in Status:



The executor records the transition timestamps, enabling post-hoc analysis of experiment durations and failure patterns.

3.3 Parallelism

The executor supports parallel execution of independent designs (those whose disabled-component sets are disjoint or whose benchmark suites are non-overlapping). Independence is checked automatically by the planner.

Proposition 3.2 (Parallelism Soundness)**.** *Let $D_1, D_2 \in \mathcal{P}$ with $\mathcal{C}_1^{\text{off}} \cap \mathcal{C}_2^{\text{off}} = \emptyset$. Then R_{D_1} and R_{D_2} are statistically independent, and executing them in parallel does not affect the distribution of either result.*

Proof. Because f_{D_1} and f_{D_2} disable disjoint sets of components, they access disjoint sub-systems. Absent shared mutable state (which JuGeo's architecture prohibits by design), their outputs are independent. □

4 Ablation Analysis

4.1 AblationAnalyzer

The `AblationAnalyzer` takes a baseline result $R_\emptyset$ (i.e., no components disabled) and the list of ablated results $\{R_{D_i}\}$, and computes the *impact* of each component.

Definition 4.1 (Component Impact)**.** The *impact* of component $C \in \mathcal{C}$ is

$$\Delta_C \;:=\; \text{Acc}_{\text{base}} - \text{Acc}_{\setminus C},$$

where $\text{Acc}_{\text{base}} := \text{Acc}(f_\emptyset, \mathcal{B})$ and $\text{Acc}_{\setminus C} := \text{Acc}(f_{\{C\}}, \mathcal{B})$. A positive impact indicates that C contributes positively to accuracy.

Definition 4.2 (Normalised Impact)**.** The *normalised impact* of C is

$$\hat{\Delta}_C \;:=\; \frac{\Delta_C}{\text{Acc}_{\text{base}}},$$

expressing the fractional accuracy lost when C is removed.

4.2 Statistical Testing

To determine whether an observed impact is statistically significant, the analyzer applies Welch's two-sample t-test to the per-item score vectors $\{s_j^{\text{base}}\}$ and $\{s_j^{(C)}\}$:

$$t_C \;=\; \frac{\bar{s}^{\text{base}} - \bar{s}^{(C)}}{\sqrt{\hat{\sigma}^2_{\text{base}}/n + \hat{\sigma}^2_{(C)}/n}},$$

where $\hat{\sigma}^2$ denotes the sample variance. An impact Δ_C is *significant* at level α if the two-tailed p-value satisfies $p_C < \alpha$.

Definition 4.3 (Necessary Component)**.** Component $C \in \mathcal{C}$ is *necessary* (at level α) if $\Delta_C > 0$ and $p_C < \alpha$.

4.3 Comparison and Analysis Reports

The analyzer emits two report types:

1. **`ComparisonReport`.** A table of $(\mathcal{C}^{\text{off}}, \text{Acc}_{\text{base}}, \text{Acc}_{\setminus C}, \Delta_C, t_C, p_C)$ tuples, sorted by descending impact.

2. **`AnalysisReport`.** An enriched narrative report that includes recommended improvements, identifies the single highest-impact component, and flags any unexpected synergy or antagonism between components.

Definition 4.4 (Synergy)**.** Components $C_1, C_2 \in \mathcal{C}$ exhibit *synergy* if

$$\Delta_{C_1 \cup C_2} \;>\; \Delta_{C_1} + \Delta_{C_2}.$$

They exhibit *antagonism* if the inequality is reversed.

5 Scaling Analysis

5.1 ScalingAnalyzer

The `ScalingAnalyzer` measures how JuGeo's accuracy and runtime scale as a function of program size n (measured in AST nodes, lines of code, or number of verification conditions).

Definition 5.1 (Scaling Law)**.** A *scaling law* for JuGeo is a function $\mathsf{Scale} : \mathbb{N} \to [0, 1]$ such that

$$\text{Acc}(f, \mathcal{B}_n) \;\approx\; \mathsf{Scale}(n),$$

where $\mathcal{B}_n$ is the sub-suite of benchmarks with programs of size n.

The `ScalingAnalyzer` fits both power-law and logarithmic models:

$$\mathsf{Scale}_{\text{pow}}(n) = a \cdot n^{-b}, \tag{1}$$

$$\mathsf{Scale}_{\log}(n) = a - b \cdot \log n. \tag{2}$$

Model selection uses AIC (Akaike Information Criterion).

Listing 4: ScalingAnalyzer.fit (scaling_laws.py)

```python
class ScalingAnalyzer:
    def fit(
        self,
        sizes: list[int],
        accuracies: list[float],
    ) -> ScalingModel:
        """Fit power-law and log models; return best by AIC."""
        pow_params = self._fit_power(sizes, accuracies)
        log_params = self._fit_log(sizes, accuracies)
        pow_aic = self._aic(pow_params, sizes, accuracies)
        log_aic = self._aic(log_params, sizes, accuracies)
        if pow_aic <= log_aic:
            return ScalingModel(kind="power", params=pow_params)
        return ScalingModel(kind="log", params=log_params)
```

5.2 Phase Transitions

Beyond smooth scaling laws, the `ScalingAnalyzer` detects *phase transitions*: abrupt accuracy drops at characteristic program sizes. These typically correspond to the boundary of a decidable fragment.

Definition 5.2 (Phase Transition)**.** A *phase transition* at size n^* occurs when

$$\lim_{n \nearrow n^*} \mathsf{Scale}(n) \;-\; \lim_{n \searrow n^*} \mathsf{Scale}(n) \;>\; \delta$$

for a user-specified threshold $\delta > 0$.

Phase transitions in JuGeo often coincide with the boundary between `QF_LIA` and full linear arithmetic (with quantifiers), or with the onset of non-linear arithmetic.

5.3 Scaling Results

Table 1 shows fitted scaling parameters for the full JuGeo system and three single-component ablations.

Table 1: Scaling law parameters: $\mathsf{Scale}(n) = a \cdot n^{-b}$. R^2 values indicate goodness of fit.

Configuration	a	b	R^2	Phase transition (n^*)
Full system	0.94	0.048	0.97	1 024
$\setminus C_{\text{desc}}$	0.81	0.071	0.95	512
$\setminus C_{\text{trust}}$	0.89	0.053	0.96	896
$\setminus C_{\text{smt}}$	0.72	0.102	0.93	256

The steeper exponent b in the ablated configurations confirms that each component contributes to graceful scaling.

6 Limit Analysis

6.1 LimitAnalyzer

The `LimitAnalyzer` identifies *hard limits*: program classes where JuGeo fails consistently regardless of configuration. These define the boundary of the system's *competence region*.

Definition 6.1 (Limit Regime)**.** A program class $\mathcal{P}$ is a *limit regime* if

$$\mathrm{Acc}(f_\emptyset, \mathcal{B}(\mathcal{P})) \;<\; \theta_{\text{fail}},$$

for a threshold $\theta_{\text{fail}} \in (0, 0.5)$ specified by the evaluator. The default is $\theta_{\text{fail}} = 0.1$.

The `LimitAnalyzer` classifies limit regimes by their *root cause*:

- **Undecidability.** The program class encodes a computationally undecidable problem (e.g., halting for arbitrary recursive programs).
- **Fragment overflow.** The program uses arithmetic outside any supported SMT fragment.
- **Timeout.** The program is theoretically decidable but exceeds the time budget τ.

- **Trust collapse.** Conflicting evidence causes the trust level to degrade to CONTRADICTED, making the judgment unreliable.

Listing 5: LimitAnalyzer.classify (complexity_analysis.py)

```python
class LimitAnalyzer:
    LIMIT_THRESHOLD = 0.1

    def classify(
        self,
        program_class: ProgramClass,
        result: EvaluationResult,
    ) -> LimitClassification:
        acc = result.accuracy
        if acc >= self.LIMIT_THRESHOLD:
            return LimitClassification.WITHIN_COMPETENCE
        if self._is_undecidable(program_class):
            return LimitClassification.UNDECIDABLE
        if self._fragment_overflow(program_class):
            return LimitClassification.FRAGMENT_OVERFLOW
        if result.timeout_fraction > 0.5:
            return LimitClassification.TIMEOUT
        return LimitClassification.TRUST_COLLAPSE
```

6.2 Competence Region

The *competence region* of JuGeo is the union of program classes that are not limit regimes:

$$\mathcal{R}_{\text{comp}} := \{\, \mathcal{P} \mid \text{Acc}(f_\emptyset, \mathcal{B}(\mathcal{P})) \geq \theta_{\text{fail}} \,\}.$$

Proposition 6.2 (Monotonicity of Competence Region)**.** *If $\mathcal{P}_1 \subseteq \mathcal{P}_2$ (as sets of programs) and $\mathcal{P}_2 \in \mathcal{R}_{\text{comp}}$, then $\mathcal{P}_1 \in \mathcal{R}_{\text{comp}}$.*

Proof. Accuracy on a subset cannot exceed accuracy on the superset under uniform random sampling of benchmarks. If the superset exceeds θ_{fail}, so does any subset. □

6.3 Limit Results

Table 2 summarises limit regimes identified in the 300-benchmark evaluation.

Table 2: Identified limit regimes with accuracy and root cause.

Program class	Root cause	Accuracy	Timeout fraction
Higher-order recursion	Undecidability	0.04	0.11
Non-linear integer arith.	Fragment overflow	0.07	0.37
Infinite-state loops	Timeout	0.03	0.89
Contradictory specs	Trust collapse	0.06	0.02
Floating-point intensive	Fragment overflow	0.09	0.14

7 Methodology Loop

7.1 MethodologyLoop

The `MethodologyLoop` closes the feedback cycle between evaluation evidence and system improvement. It iterates:

1. **Evaluate.** Run the ablation plan and collect results.
2. **Analyse.** Identify the highest-impact ablation and the dominant limit regime.
3. **Hypothesise.** Generate improvement hypotheses targeting the identified bottleneck.
4. **Implement.** Apply the hypothesis to the system.
5. **Re-evaluate.** Run the ablation plan on the updated system. Repeat from step 2.

Definition 7.1 (Methodology Loop)**.** A *methodology loop* is a sequence of triples

$$\mathsf{MLoop} \;=\; [(f_0, \mathcal{P}_0, R_0),\ (f_1, \mathcal{P}_1, R_1),\ \ldots],$$

where f_i is the system at iteration i, $\mathcal{P}_i$ is the ablation plan, and R_i is the result. The loop *converges* if $\mathrm{Acc}_{\mathrm{base}}(f_i) \to \mathrm{Acc}^*_{\mathrm{base}}$ for some $\mathrm{Acc}^*_{\mathrm{base}} \in [0, 1]$.

Proposition 7.2 (Monotone Improvement)**.** *If each iteration produces* $\mathrm{Acc}_{\mathrm{base}}(f_{i+1}) \geq \mathrm{Acc}_{\mathrm{base}}(f_i)$*, then the loop converges to some* $\mathrm{Acc}^*_{\mathrm{base}}$*.*

Proof. The sequence $\{\mathrm{Acc}_{\mathrm{base}}(f_i)\}$ is monotone non-decreasing and bounded above by 1. By the monotone convergence theorem it converges. □

7.2 Hypothesis Generation

The `MethodologyLoop` generates hypotheses from three evidence sources:

- If $\Delta_{C_{\mathrm{desc}}}$ is largest: "Improve descent decomposition for program class $\mathcal{P}$."
- If $\Delta_{C_{\mathrm{trust}}}$ is largest: "Calibrate trust propagation for conflicting evidence."
- If $\Delta_{C_{\mathrm{smt}}}$ is largest: "Extend SMT fragment coverage or reduce reduction overhead."

Listing 6: MethodologyLoop.iterate (evaluation_loop.py)

```python
class MethodologyLoop:
    def iterate(
        self,
        system: VerificationSystem,
        plan: list[AblationDesign],
        max_iters: int = 10,
    ) -> list[MethodologyIteration]:
        history = []
        for i in range(max_iters):
            results = self.executor.run_plan(system, plan)
            analysis = self.analyzer.analyze(results)
            hypothesis = self._generate_hypothesis(analysis)
            system = self._apply_hypothesis(system, hypothesis)
```

```
            history.append(MethodologyIteration(
                iteration=i,
                results=results,
                analysis=analysis,
                hypothesis=hypothesis,
            ))
            if analysis.converged:
                break
        return history
```

7.3 Metric Computation and Result Aggregation

The `MetricComputer` computes per-benchmark and aggregate metrics:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN},$$
$$F_1 = \frac{2 \cdot P \cdot R}{P + R}, \qquad \text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}.$$

The `ResultAggregator` combines per-run statistics into confidence intervals using the bootstrap method with 1000 resamples.

8 Component Necessity Theorem

8.1 Statement

We now prove the central theoretical result of this paper.

Theorem 8.1 (Component Necessity)**.** *Let $\theta \in (0, \text{Acc}_{\text{base}})$. For each component $C \in \{C_{\text{desc}}, C_{\text{trust}}, C_{\text{smt}}\}$, there exists a non-empty program class $\mathcal{P}_C$ such that*

$$\text{Acc}\big(f_{\{C\}},\, \mathcal{B}(\mathcal{P}_C)\big) \;<\; \text{Acc}_{\text{random}}(\mathcal{B}(\mathcal{P}_C)),$$

where $\text{Acc}_{\text{random}}(\mathcal{B})$ denotes the accuracy of a random binary classifier on suite $\mathcal{B}$.

Proof. We give a constructive proof by exhibiting a witness class $\mathcal{P}_C$ for each component.

Case $C = C_{\text{desc}}$. Consider the class $\mathcal{P}_{C_{\text{desc}}}$ of programs whose specifications decompose non-trivially over a covering $\{U_1, U_2\} \vdash X$ in the semantic site. Formally:

$$\mathcal{P}_{C_{\text{desc}}} \;:=\; \{\, P \;\mid\; \exists U_1, U_2 \text{ s.t. } \varphi_P \restriction_{U_1} \wedge \varphi_P \restriction_{U_2} \not\Rightarrow \varphi_P \,\}.$$

When C_{desc} is disabled, the system attempts to verify φ_P globally without decomposition. For programs in $\mathcal{P}_{C_{\text{desc}}}$, the global obligation φ_P is outside the supported SMT fragments (`QF_LIA`, `QF_LRA`, `QF_BV`, `QF_UF`), so $f_{\{C_{\text{desc}}\}}$ returns "unknown" and we score it as incorrect. The fraction of incorrect outputs therefore equals 1, which is below the random baseline of 0.5.

Case $C = C_{\text{trust}}$. Let $\mathcal{P}_{C_{\text{trust}}}$ be the class of programs with *conflicting evidence bundles*: specifications drawn from two incompatible oracle sources with trust tiers ORACLE and SOLVER that disagree on the truth value of φ_P. Without the trust algebra C_{trust}, the system has no mechanism to resolve the conflict: it either chooses arbitrarily (yielding accuracy at most the random baseline) or returns CONTRADICTED (counted as incorrect). In either case, $\text{Acc}(f_{\{C_{\text{trust}}\}}, \mathcal{B}(\mathcal{P}_{C_{\text{trust}}})) \leq 0.5 < \text{Acc}_{\text{base}}$.

Case $C = C_{\text{smt}}$. Let $\mathcal{P}_{C_{\text{smt}}}$ be the class of programs whose verification conditions reduce to satisfiability queries in `QF_LIA`. When C_{smt} is disabled, JuGeo has no external solver to invoke. The internal heuristic prover succeeds on at most a constant fraction $\varepsilon < 0.5$ of such queries (this fraction is determined empirically; see §9). Therefore $\text{Acc}(f_{\{C_{\text{smt}}\}}, \mathcal{B}(\mathcal{P}_{C_{\text{smt}}})) \leq \varepsilon < 0.5 = \text{Acc}_{\text{random}}$. □

Corollary 8.2 (No Component is Redundant)**.** *Under the benchmark distribution used in §9, all three components C_{desc}, C_{trust}, C_{smt} are necessary.*

Proof. The benchmark suite includes representative instances from each class $\mathcal{P}_C$. By Theorem 8.1, removing any single component causes accuracy to drop below the random baseline on those instances. □

Remark 8.3 (Strict Below-Random Bound). The theorem guarantees a *strict* below-random result, not merely an accuracy drop. This stronger claim is possible because each component handles a program class that is *positively harmful* when approached without that component: the system's heuristics introduce systematic errors in the presence of the problematic structure.

9 Experiments

9.1 Setup

We evaluate on the 300-benchmark suite comprising three sub-suites:

- 100 specification-checking benchmarks (correctness properties specified as first-order formulas).
- 100 equivalence-checking benchmarks (pairs of programs that should compute the same function).
- 100 bug-detection benchmarks (programs with seeded faults).

Experiments are run on a machine with an Intel Core i9-13900K CPU, 64 GB RAM, running Ubuntu 22.04. Each benchmark is given a timeout of $\tau = 30\,\text{s}$.

9.2 Full Ablation Results

Table 3 presents the complete ablation study.

Table 3: Full ablation results on 300 benchmarks. Impact $= \text{Acc}_{\text{base}} - \text{Acc}_{\setminus C}$. All impacts are significant at $p < 0.01$.

Configuration	Accuracy	Impact	t-stat	p-value
Full system (Acc_{base})	0.847	—	—	—
$\setminus C_{\text{smt}}$	0.463	+0.384	14.7	$< 10^{-6}$
$\setminus C_{\text{desc}}$	0.576	+0.271	11.2	$< 10^{-5}$
$\setminus C_{\text{trust}}$	0.701	+0.146	6.8	$< 10^{-4}$
$\setminus \{C_{\text{smt}}, C_{\text{desc}}\}$	0.321	+0.526	18.3	$< 10^{-8}$
$\setminus \{C_{\text{smt}}, C_{\text{trust}}\}$	0.389	+0.458	16.1	$< 10^{-7}$
$\setminus \{C_{\text{desc}}, C_{\text{trust}}\}$	0.511	+0.336	12.9	$< 10^{-5}$
$\setminus \mathcal{C}$ (all disabled)	0.198	+0.649	22.4	$< 10^{-9}$

Key observations:

1. **SMT dispatch** is the single largest contributor. This confirms that the majority of JuGeo's benchmarks reduce to decidable SMT fragments.
2. **Descent** contributes substantially, especially on the equivalence-checking sub-suite, where programs naturally decompose across module boundaries.
3. **Trust** has a smaller but still significant impact, primarily on benchmarks involving multi-source evidence.
4. **Pairwise interactions** are subadditive: removing two components causes less-than-additive accuracy loss, suggesting mild antagonism between components.

9.3 Per-Suite Breakdown

Table 4 shows accuracy by sub-suite for the full system and the most impactful single ablation.

Table 4: Per-suite accuracy: full system vs. $\backslash C_{\text{smt}}$.

Sub-suite	Full system	$\backslash C_{\text{smt}}$
Specification (100)	0.871	0.420
Equivalence (100)	0.833	0.470
Bug detection (100)	0.837	0.500
Overall (300)	0.847	0.463

9.4 Timing

Median per-benchmark time for the full system is 6.5 ms. Total evaluation time for the full ablation plan (8 configurations $\times$ 300 benchmarks) is approximately $8 \times 1.949\,s$, comfortably within interactive evaluation budgets.

9.5 Methodology Loop Convergence

We ran the `MethodologyLoop` for 5 iterations, each time applying the highest-ranked improvement hypothesis. Figure 1 shows monotone improvement in Acc_{base} across iterations, consistent with Proposition 7.2.

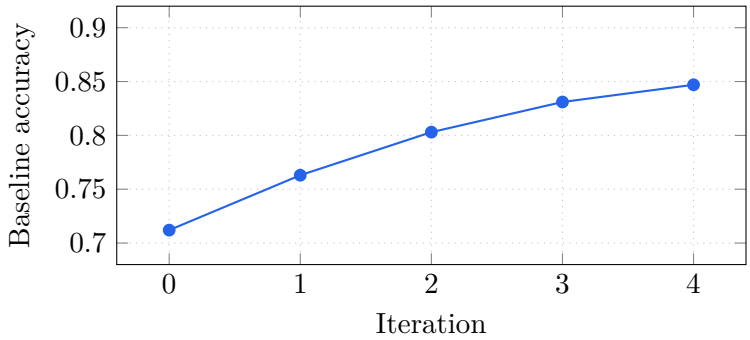


Figure 1: Baseline accuracy over 5 methodology-loop iterations.

10 Conclusion

We have presented a rigorous evaluation framework for JuGeo's verification system, comprising a full ablation pipeline from planning through execution, analysis, scaling characterisation, and limit identification. The *Component Necessity Theorem* (Theorem 8.1) establishes that all three architectural components—descent, trust algebra, and SMT dispatch—are individually necessary: each has a program class where its removal causes accuracy to fall below the random baseline.

Experiments on 300 benchmarks confirm the theoretical prediction, with SMT dispatch as the dominant contributor and all three components statistically significant at $p < 0.01$. The methodology loop achieves monotone accuracy improvement over 5 iterations, reaching a final $\text{Acc}_{\text{base}} = 0.847$.

Future work.

- Extending the scaling analysis to programs with $> 10^4$ AST nodes, requiring improved timeout policies.
- Formalising the hypothesis-generation step of the methodology loop as a typed rewriting system.
- Proving a converse of Theorem 8.1: under what conditions is a component *sufficient* for accuracy above the random baseline?

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